

Fourier Series

used for periodic signals to expand them in harmonics

$$x(t \pm T_0) = x(t)$$

* Dirichlet condition $\rightarrow$ for Fourier to exist

$$\text{Harmonic} = \frac{\text{cycles}}{\text{sec}}$$

$$x(t) = 2 \sin \omega t + \sin 2\omega t + 7 \sin 3\omega t \dots$$

$\uparrow$ even harmonic
 $\downarrow$ odd harmonic

$\left(\text{fundamental } (\omega) = 2\pi f \right)$
 $\rightarrow$ fundamental harmonic

Conditions for existence

- C1: finite no. of max, min over time period
- C2: finite no. of discontinuity
- C3: absolutely integrable over time period T

Trigonometric values

$\rightarrow$ dc value of $x(t)$

$$x(t) = \text{dc of } x(t) + \text{Sine} + \text{Cosine}$$

$$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega t + \sum_{n=1}^{\infty} b_n \sin n\omega t$$

$$a_0 = \frac{\int_0^{T_0} x(t) \cdot dt}{T_0}$$

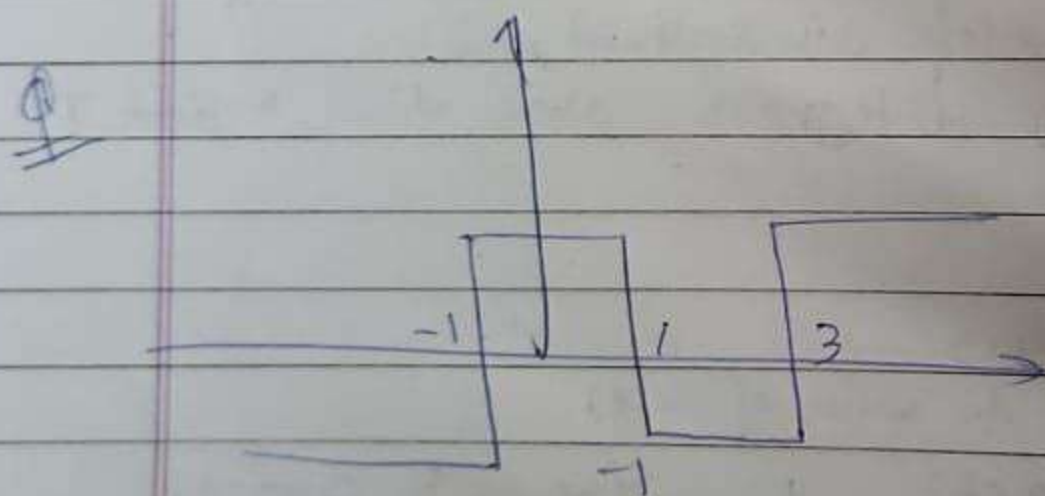
$$a_n = \frac{2}{T_0} \int_{T_0} x(t) \cos n \omega t dt$$

$$b_n = \frac{2}{T_0} \int_{T_0} x(t) \sin n \omega t dt$$

Ex. $x(t) = 4 + 3 \cos 2\omega t + 4 \cos 3\omega t$ $\rightarrow 5 \sin 2\omega t \rightarrow 2^{\text{nd}}$
 $\rightarrow 2 \sin 3\omega t \rightarrow 3^{\text{rd}}$

o $x(t) \rightarrow$ sym about t axis $\Rightarrow a_0 = 0$ (Intg = 0)

o $x(t) \rightarrow$ even $\rightarrow b_n = 0$
 o $x(t) \rightarrow$ odd $\rightarrow a_n = 0$ $\rightarrow$ Common rule



$$a_0 = 0$$

sym about

$$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos n \omega t + \sum_{n=1}^{\infty} b_n \sin n \omega t$$

$b_n = 0$ (even signal)

$$a_n = \frac{2}{T_0} \int_{-1}^3 x(t) \cos n \omega t dt$$

$$T_0 = 4$$

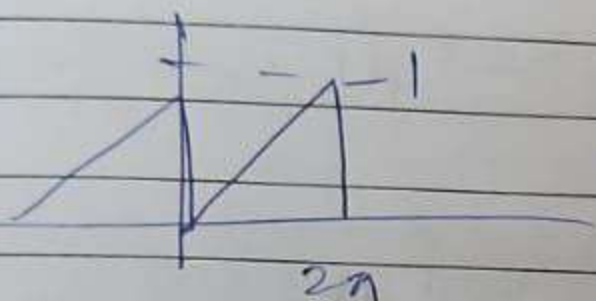
$$\frac{2\pi}{T_0} = \omega_0$$

$$\left(\frac{\pi}{2} = \omega_0 \right)$$

$$a_n = \frac{2}{2\pi} \int_{-1}^1 x(t) \cos\left(n \frac{2\pi}{2} t\right) dt$$

now we will see value of $x(t)$

$$\therefore \int_{-1}^1 x(t) = 1 \quad \int_{-1}^1 x(t) = -1$$



$$x(t) = a_0 + \sum a_n \cos n\omega_0 t + \sum b_n \sin n\omega_0 t$$

$$a_0 = \frac{1}{2\pi} \int_0^{2\pi} x(t) dt = \frac{1}{2} \times 2\pi \times 1 = \frac{1}{2}$$

$$\begin{aligned} a_n &= \frac{2}{T_0} \int_0^{2\pi} x(t) \cos(n\omega_0 t) dt \\ &= \frac{1}{\pi} \int_0^{2\pi} x(t) \cos(nt) dt \\ &= \frac{1}{\pi} \int_0^{2\pi} \frac{1}{2\pi} \cos(nt) dt \end{aligned}$$

$$\omega_0 = \frac{2\pi}{T_0}$$

$$= \frac{1}{2\pi^2} \int_0^{2\pi} t \cos nt dt$$

$$= \left(\frac{t \sin nt}{n} + \frac{\cos nt}{n^2} \right) \frac{1}{n}$$

$$= \left(0 + \left(\frac{\cos 2\pi n}{n} \right) \frac{1}{n} \right) - \frac{1}{n^2}$$

⇒ Complex Exponential Fourier series

$$x(t) = \sum_{n=-\infty}^{\infty} C_n \cdot e^{jn\omega_0 t} \rightarrow \text{+ve (exp)}$$

$$C_n = \frac{1}{T_0} \int_{T_0} x(t) e^{-jn\omega_0 t} dt \quad \text{--- (1)}$$

Is -ve exp

⇒ a_n, b_n, C_n (Relation)

$$a_n = \frac{1}{T_0} \int_{T_0} x(t) e^{jn\omega_0 t} dt \rightarrow \text{Parasol op}$$

$$C_{-n}^* = \frac{1}{T_0} \int_{T_0} x^*(t) e^{-jn\omega_0 t} dt \rightarrow \text{Conjugate op}$$

$$C_n = C_{-n}^* \Rightarrow x(t) = x^*(t) \Rightarrow \text{Real}$$

↳ Comparing

↓
Purely
Real signal

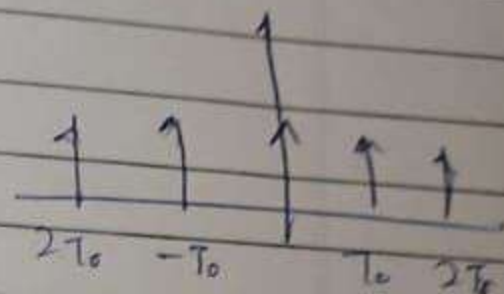
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$$x(t) = \int C_n e^{jn\omega_0 t}$$

1. Fourier series represented by impulse train

$$s(t) = \sum_{-\infty}^{\infty} c_n e^{jn\omega_0 t}$$

$$c_n = \frac{1}{T_0} \int_{T_0} s(t) \cdot e^{-jn\omega_0 t} dt$$



$$c_n = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} s(t) e^{-jn\omega_0 t} dt$$

$$x(t) * s(t - t_0) = s(t - t_0) \cdot x(t)$$

$$c_n = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} s(t) dt \Rightarrow$$

$$c_n = \frac{1}{T_0}$$

$$\begin{aligned} e^{-j\alpha} + e^{j\alpha} &= 2\cos\alpha \\ e^{j\alpha} - e^{-j\alpha} &= 2j\sin\alpha \end{aligned} \rightarrow \text{for conversion}$$

Properties of Fourier Series

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1) Linearity

$$x_1(t) \rightarrow a_n$$

$$x_2(t) \rightarrow c_{2n}$$

$$\alpha x_1(t) + \beta x_2(t) = x(t) \rightarrow c_n$$

$$c_n = \frac{1}{T_0} \int_{T_0} x(t) \cdot e^{-j\omega_0 n t} dt$$

(limit is T_0 of c_n)

$$c_n = \alpha a_n + \beta c_{2n}$$

2) Conjugation

$$x(t) \Rightarrow c_n$$

Take conj both sides

$$c_n^* = \frac{1}{T_0} \int_{T_0} x^*(t) \cdot e^{j\omega_0 n t} dt$$

$$\underline{n \rightarrow -n}$$

$$(x^*(t) \rightarrow \underline{c_n^*})$$

3) Time reversal

$$(t \rightarrow -t)$$

$$c_n \longleftrightarrow c_{-n}$$

4. Time scaling

$$C_n = \frac{1}{T_0} \int_0^{T_0} x(t) e^{-j\omega_n t} dt$$

$$x(t) \rightarrow x(at) ; |a| > 1$$

$$T_0' \rightarrow \frac{T_0}{a}$$

$$\omega_0 \rightarrow \frac{2\pi a}{T_0}$$

$$(C_n' = C_n) \text{ No change just period} = \frac{T_0}{a}$$

5. Time shifting

$$x(t) \rightarrow x(t - t_0)$$

$$C_n' = e^{-j\omega_n t_0} C_n$$

$$x(t) \rightarrow x(t + t_0)$$

$$C_n' = e^{j\omega_n t_0} C_n$$

6. Frequency shifting

$$C_{n-m} \rightarrow \text{New } x(t) \rightarrow e^{j\omega_0 t}$$

7. Convolution in time

$$x_1(t) \rightarrow C_m$$

$$x_2(t) \rightarrow C_n$$

$$x_1(t) * x_2(t) \rightarrow T_o (C_m \cdot C_n)$$

8. Multiplication in time

$$x_1(t) \cdot x_2(t) \rightleftharpoons C_m + C_n$$

↳ discrete

9. Diff

$$\frac{d x(t)}{dt} \rightarrow j\omega_0 C_n$$

$$\int x(t) dt \rightarrow \frac{C_n}{j\omega_0} \quad (\text{Integ})$$

$$-2ee^{-}$$

$$\frac{2\pi}{T} = \omega$$

$$\frac{2\pi}{\omega} = T$$

⇒ Fourier series for discrete systems

$$x(n) = \sum_{k=-N}^N x_k e^{j\omega_0 n k}$$

$$x_n = \frac{1}{N} \sum x(n) e^{-j\omega_0 n k}$$

Magnitude spectrum

$$|X_n| \text{ vs } \omega$$

$$\text{or } x_n \text{ vs } \omega$$

Phase Spectrum

⇒

Phase of

x_k

vs

ω

vs ω

Fourier Transform

- ↳ Analysis of frequency.
- ↳ exists for ES, PS and Impulse related signals.

$$x(t) \xLeftrightarrow \quad \begin{array}{c} X(j\omega) \text{ or } X(f) \\ \downarrow \quad \downarrow \\ \omega \quad \text{Hz} \end{array}$$

Formulae

1) $x(t) \rightarrow X(j\omega)$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt \quad \leftarrow \text{FT}$$

2) $X(j\omega) \rightarrow x(t)$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega \quad \leftarrow \text{IFT}$$

✱ only for absolutely integrable program.

→ Condition for existence of fourier transform

0 Finite number of maxima & minima over a finite period

0 " discontinuities "

0 Absolutely integrable over $-\infty$ to ∞

↳ ES, impulse related signals.

$$x(t) = a(t)$$

To not absolutely only

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt$$

$$= \int_{-\infty}^{\infty} a(t) \cdot e^{-j\omega t} dt$$

$$X(j\omega) = \int_0^{\infty} e^{-j\omega t} dt$$

$$= \frac{-1}{j\omega} \left[e^{-j\omega t} \right]_0^{\infty}$$

$$X(j\omega) = \frac{1}{j\omega}$$

($e^{-j\omega \infty}$)
 $\hookrightarrow$ ND

$$X(j\omega) = \frac{1}{j\omega} + \pi \delta(\omega)$$

Properties

1) Linearity

$$\alpha x_1(t) \implies \alpha X_1(j\omega)$$

$$\beta x_2(t) \implies \beta X_2(j\omega)$$

$$\alpha x_1(t) + \beta x_2(t) \implies \alpha X_1(j\omega) + \beta X_2(j\omega)$$

2)

Conjugation

$$\begin{aligned} x(t) &\xrightarrow{\quad} X(j\omega) \\ x^*(t) &\xrightarrow{\quad} X^*(-j\omega) \end{aligned}$$

3)

Area under $x(t)$: $\int_{-\infty}^{\infty} x(t) \cdot dt$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} \cdot dt$$

$$\downarrow \omega = 0$$

$$X(0) = \int_{-\infty}^{\infty} x(t) \cdot dt = \text{area under time domain}$$

4)

Area under $X(j\omega)$: $\int_{-\infty}^{\infty} X(j\omega) \cdot d\omega$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$$

=0)

$$x(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) d\omega = \text{area}$$

5) Time Reversal

$$\begin{aligned} x(t) &\xrightarrow{\quad} X(j\omega) \\ x(t) &\xrightarrow{\quad} X(-j\omega) \end{aligned}$$

for proof:

$t \rightarrow -t$ but only for $x(t)$
 $\therefore \downarrow$

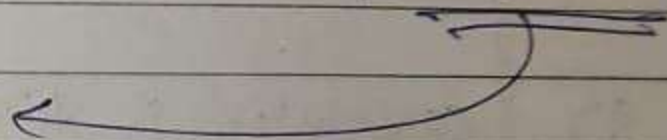
$x(-t)$

$$\therefore X(j\omega) = \int_{-\infty}^{\infty} x(-t) e^{-j\omega t} dt$$

$-t = z \rightarrow$ for standard formula.

$$\therefore e^{j\omega z} \equiv e^{-j\omega(-z)}$$

$$\therefore \boxed{X(-j\omega)}$$



6) Time scaling

$$x(at) \xrightarrow{\quad} \frac{1}{|a|} X\left(\frac{j\omega}{a}\right)$$

7)

Time shifting

$$x(t) \xLeftrightarrow{\quad} X(j\omega)$$

$$x(t \pm t_0) \xLeftrightarrow{\quad} X(j\omega) \cdot e^{\pm j\omega t_0}$$

8)

Frequency shifting

$$e^{\pm j\omega_0 t} \cdot x(t) \xLeftrightarrow{\quad} X[j(\omega \mp \omega_0)]$$

++ case, here the sign is opposite.

9)

Convolution in time

$$x_1(t)$$

$$x_2(t)$$

$$x_1(t) * x_2(t) \xLeftrightarrow{\quad} X_1(j\omega) \cdot X_2(j\omega)$$

10) Multiplication in time

[condition in freq]

$$x_1(t) \Leftrightarrow X_1(j\omega)$$

$$x_2(t) \Leftrightarrow X_2(j\omega)$$

$$x_1(t) \cdot x_2(t) \Leftrightarrow \frac{1}{2\pi} (X_1(j\omega) * X_2(j\omega))$$

Proof with inverse FT of
RHS.

Diff in time

$$x(t) \Rightarrow X(j\omega)$$

$$x'(t) \Rightarrow j\omega X(j\omega)$$

so on

Intg in time

$$x(t) \Leftrightarrow X(j\omega)$$

$$t \int_{-\infty}^{\infty} x(\tau) d\tau \Leftrightarrow \frac{X(j\omega)}{j\omega} + \pi X(0) \delta(\omega)$$

Proof : $\underline{\underline{u(t) * x(t) = \int_{-\infty}^{\infty} x(\tau) u(t-\tau) d\tau}}$

B) Diff. in frequency

$$t. x(t) \Rightarrow j \frac{dX(j\omega)}{d\omega}$$

$\therefore$

$$t^n x(t) \Rightarrow j^n \frac{d^n X(j\omega)}{d\omega^n}$$

14) Modulation

$$x(t) \Rightarrow X(j\omega) \text{ or } X(\omega)$$

$$(i) x(t) \cdot \cos \omega_0 t \Rightarrow \frac{1}{2} [X(\omega + \omega_0) + X(\omega - \omega_0)]$$

$$(ii) x(t) \cdot \sin \omega_0 t \Rightarrow \frac{j}{2} [X(\omega + \omega_0) - X(\omega - \omega_0)]$$

$e^{j\omega t}$
 $e^{-j\omega t}$



$$2\pi \delta(\omega - \omega_0)$$

$$2\pi \delta(\omega + \omega_0)$$

13)

Parsavals Power Theorem

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$$x(t) \rightleftharpoons c_n \quad (\text{coeff for complex fourier series})$$

$$P_x(t) = \sum_{n=-\infty}^{\infty} |c_n|^2$$

Result

$$x(t) \rightleftharpoons x(j\omega)$$

$$E_x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} |x(j\omega)|^2 d\omega$$

Important result $\Rightarrow |x(t)|^2 = x(t) \cdot \underset{\substack{\downarrow \\ \text{conjugate}}}{x^*(t)}$

$$E_x(t) = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$= \int_{-\infty}^{\infty} x(t) \cdot x^*(t) dt$$

$$= \int_{-\infty}^{\infty} x(t) \left(\underbrace{\frac{1}{2\pi} \int_{-\infty}^{\infty} x^*(j\omega) e^{-j\omega t} d\omega}_{\substack{\downarrow \\ \text{FT}}} \right) dt$$

$$E_x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} x(j\omega) \cdot x^*(j\omega) d\omega$$

$$E_x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} |x(j\omega)|^2 d\omega$$

HP

16)

Duality property

$$x(t) \rightleftharpoons X(\omega)$$

$$X(t) \rightleftharpoons 2\pi x(-\omega)$$

Proof:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega) e^{j\omega t} d\omega$$

$$t \rightarrow -t \quad (\text{not substitution})$$

$$2\pi x(-t) = \int_{-\infty}^{\infty} X(j\omega) e^{-j\omega t} d\omega$$

$$t \rightarrow \omega, \quad \omega \rightarrow t$$

$$2\pi x(-\omega) = \int_{-\infty}^{\infty} X(jt) e^{-j\omega t} dt$$

FT

FT

Signal

→ Fourier transform of basic signals

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1) DC value

$$x(t) = A_0$$

→ not absolutely intg.

$$\int_{-\infty}^{\infty} |x(t)| dt < \infty$$

$$\int_{-\infty}^{\infty} A_0 dt$$

$$A_0 \int_{-\infty}^{\infty} 1 dt$$

$$A_0 \times \infty = \infty$$

→ Not an absolutely signal

$$X(\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt$$

$$X(\omega) = A_0 \int_{-\infty}^{\infty} e^{-j\omega t} dt$$

$$= -\frac{A_0}{j\omega} \left[e^{-j\omega t} \right]_{-\infty}^{\infty}$$

$$= -\frac{A_0}{j\omega} \left[e^{-\infty j} - e^{\infty j} \right]$$

$$\boxed{X(\omega) = \frac{A_0}{\omega} 2 \sin(\infty)} \rightarrow \text{Undefined.}$$

$$x(t) \xrightarrow{FT} X(\omega) = A_0 \delta(\omega)$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} A_0 \delta(\omega) \cdot e^{j\omega t} d\omega$$

$$= \frac{A_0}{2\pi} \int_{-\infty}^{\infty} \delta(\omega) \cdot e^{j\omega t} d\omega$$

$$\left[\delta(\omega - 0) e^{j\omega t} \right] \rightarrow \text{substitution}$$

Ing
Prof

$$x(t) = \frac{A_0}{2\pi}$$

$$\Leftrightarrow \frac{A_0}{2\pi} \Leftrightarrow A_0 \delta(\omega)$$

$$\left[\begin{array}{ccc} \frac{A_0}{2\pi} & \xrightarrow{\quad} & A_0 \delta(\omega) \\ A_0 & \xrightarrow{\quad} & 2\pi A_0 \delta(\omega) \end{array} \right]$$

Domain: time frequency

2) Impulse signal

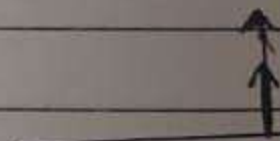
$$x(t) = \delta(t)$$

$$X(\omega) = \int_{-\infty}^{\infty} \delta(t) \cdot e^{-j\omega t} \cdot dt$$

$\downarrow$
 $\delta(t-0)$

$$X(\omega) = \int_{-\infty}^{\infty} \delta(t) \cdot 1 \cdot dt$$

$$\boxed{X(\omega) = 1}$$



3) Exponential signals.

(i) $x(t) = e^{-at} u(t)$

$$X(\omega) = \int_{-\infty}^{\infty} e^{-at} u(t) \cdot e^{-j\omega t} dt$$

$$X(\omega) = \int_{0}^{\infty} e^{-j\omega t} \cdot e^{-at} dt$$

$$X(\omega) = \int_0^{\infty} e^{-t(a+j\omega)} \cdot dt$$

$$\boxed{X(\omega) = \frac{1}{a+j\omega}} \quad (a > 0)$$

(ii) $x(t) = e^{at} \cdot u(t) \quad (a > 0)$

$$X(\omega) = \int_{-\infty}^{\infty} e^{at} \cdot u(t) \cdot e^{-j\omega t} dt$$

$$= \int_{-\infty}^0 e^{at} \cdot e^{-j\omega t} dt$$

$$= \int_{-\infty}^0 e^{-(a+j\omega)t} \cdot dt$$

$$= \frac{-1}{j\omega - a} \left(1 \right)$$

$$\boxed{X(\omega) = \frac{1}{a-j\omega}} \quad (a > 0)$$

(iii) $x(t) = e^{-a|t|}; \quad (a > 0)$

$$X(\omega) = \int_{-\infty}^{\infty} e^{-a|t|} \cdot e^{-j\omega t} \cdot dt$$

$$|t| = -t \quad (-\infty \rightarrow 0), \quad +t \quad 0 \rightarrow \infty$$

$$\boxed{X(\omega) = \frac{2a}{a^2 + \omega^2}} \quad (a > 0)$$

4)

Signum fn

$$x(t) = \begin{cases} -1 & , t < 0 \\ 0 & , t = 0 \\ 1 & , t > 0 \end{cases}$$

$$X(\omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\omega t} dt$$

for FT $\rightarrow$ absolutely intg

$$\int_{-\infty}^{\infty} | \text{sgn} | = \infty$$

We can write $\boxed{\text{sgn}(t) = u(t) - u(-t)}$

$$u(t) = \lim_{a \rightarrow 0} e^{-at} u(t)$$

$$u(-t) = \lim_{a \rightarrow 0} e^{at} u(-t)$$

$$\text{sgn}(t) = \lim_{a \rightarrow 0} (e^{-at} u(t) - e^{at} u(-t))$$

$$X(\omega) = \lim_{a \rightarrow 0} \left[\frac{1}{a + j\omega} - \frac{1}{a - j\omega} \right]$$

$$X(\omega) = \lim_{a \rightarrow 0} \left(\frac{-2j\omega}{a^2 + \omega^2} \right)$$

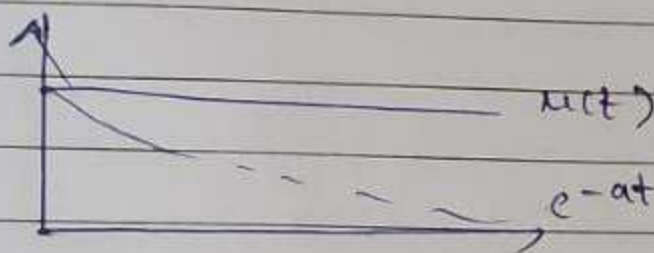
$$X(\omega) = \frac{-2j}{\omega}$$

$$= \frac{2}{j\omega}$$

$$X(\omega) = \frac{2}{j\omega}$$

5) step signal

$$x(t) = u(t) = \begin{cases} 0 & , t \leq 0 \\ 1 & , t > 0 \end{cases}$$

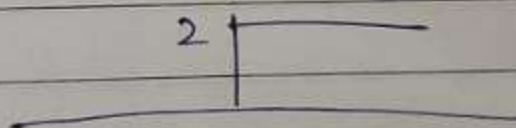


$$x(t) = \lim_{a \rightarrow \infty} e^{-at} u(t) = u(t)$$

We'll use $\text{sgn}(t)$



$$1 + \text{sgn}(t)$$



$$\frac{1 + \text{sgn}(t)}{2} = u(t)$$

$$\therefore u(t) = \frac{1}{2} + \frac{\text{sgn}(t)}{2}$$

$$\text{FT}(u(t)) = \text{FT}\left(\frac{1}{2}\right) + \text{FT}\left(\frac{\text{sgn}(t)}{2}\right)$$

TKR 6.9, 5.6

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$$X(\omega) = \pi \delta(\omega) + \frac{1}{j\omega}$$

for $u(t)$

6) $x(t) = e^{j\omega_0 t}$

$$X(\omega) = \int_{-\infty}^{\infty} e^{j\omega_0 t} \cdot e^{-j\omega t} \cdot dt$$

$\rightarrow$ conventional way

$$F\{e^{j\omega_0 t}\} = \delta(\omega - \omega_0)$$